MAS · Functions and Graphs — Patterns and Solution Strategies

Basis: 31 past exam questions from 2016–2025, marking key + exam reports. See `topics/02_Functions_Graphs.pdf` for original questions. Mainly CF (when it is the main topic, CF 21 questions, CA 7 questions). Review status: has undergone one round of independent review and been revised according to feedback (see `Review_log.md`).

Marking-key reminders

- Composite function domains are the most frequently identified source of lost marks in this topic: named in the 2016, 2017, 2022, 2023, 2024 and 2025 reports (in 2018 it was instead listed as well answered). The 2017 report identifies the cause as reluctance to set them out and inability to solve inequalities.
- Do not use "it" when explaining (2018, 2019, 2021, 2022, 2024 reports): "g is not one-to-one because g(1) = g(−1) = 1", not "it is many to one".
- To find the range, you can use the graph of the reciprocal function (the 2019 report suggests: draw the graph of 1/f first, then read off the range).
- Antiderivatives involving ln must include absolute value (2022 report; e.g. ln|x − 3| in 2020F-Q6(d)).

Pattern 1: Expressions, domains and ranges of composite functions ★★

Key principle: f(g(x)) is defined ⇔ ① x is in the domain of g, and ② g(x) lies in the domain of f.

Standard process

1. Write f(g(x)); after substituting, do not simplify first, otherwise you may lose restrictions.
2. List two sets of conditions: restrictions from g itself (denominator ≠ 0, expression under a square root ≥ 0, square root in the denominator > 0, argument of ln > 0; e.g. $y = 1/\sqrt{f(x)}$ in 2025F-Q2(a) requires f(x) > 0), and the inequality obtained from "g(x) ∈ dom f".
3. Solve the inequalities and take the intersection.

Examples

- 2017F-Q4: f(x) = 1 − √(x − 4), g(x) = 1/x² ⇒ f(g(x)) = 1 − √(1/x² − 4). Conditions: x ≠ 0 and 1/x² ≥ 4 ⇒ x² ≤ ¼ ⇒ −½ ≤ x ≤ ½, x ≠ 0.
- 2022F-Q1: f(x) = √(4 − x), g(x) = 1/x² ⇒ 4 − 1/x² ≥ 0 ⇒ x² ≥ ¼ ⇒ x ≤ −½ or x ≥ ½.
- 2016F-Q1: g∘f = 1/ln x ⇒ domain x > 0 and x ≠ 1; range y ≠ 0.
- 2019F-Q4(d): f(h(g(x))) = 1/(√(x²) − 1) = 1/(|x| − 1), not equal to f(x), because √(x²) = |x|.

Marking points: usually 1 mark for each restriction (e.g. "states x ≠ 0", "states −½ ≤ x ≤ ½"). Past questions: 2016F-Q1, 2017F-Q4(c)(d), 2018F-Q1, 2019F-Q4, 2021F-Q2(d)(e), 2022F-Q1, 2023F-Q4(b)(c), 2024F-Q2(a), 2025F-Q2(a).

Pattern 2: Inverse functions (existence, expression, graph)

Standard process

1. Existence: f must be one-to-one; the horizontal line test can be used. If not, restrict the domain; "largest possible domain" goes up to the vertex, e.g. x ≤ 1 (2016F-Q8(c)).
2. Find the expression: swap x and y (the key awards 1 mark specifically for "interchanges x, y") → solve for y → choose the positive or negative sign according to ran f^{-1} = dom f (i.e. the restriction on the original function's domain), with dom f^{-1} = ran f (2016F-Q8(d) key: $y - 1 = -\sqrt{(x + 4)}$ since $R_{f^{-1}} = D_f$; the 2016 report lists this step as an area of concern).
3. Domain and range swap: dom f^{-1} = ran f.
4. Sketching: symmetric about y = x; mark corresponding endpoints and intercepts, e.g. (a, b) ↔ (b, a).
5. "Is $f^{-1}(-1) = 4$ correct?": check whether −1 is in the range of f. f(x) = √(x − 3) ≥ 0, so $f^{-1}(-1)$ is not

defined (2018F-Q1(c)).

Inverse trigonometric functions (2020F-Q6, 2024F-Q8): $y = 2\tan x \Rightarrow f^{-1}(x) = \tan^{-1}(x/2)$. To differentiate, use implicit differentiation: $x = 2\tan y \Rightarrow 1 = 2\sec^2 y \cdot y' \Rightarrow y' = 1/(2(1 + \tan^2 y)) = 2/(4 + x^2)$.

Past questions: 2016F-Q8(c)(d), 2017F-Q4(a)(b), 2019F-Q5, 2019F-Q7, 2020F-Q6, 2021F-Q2, 2023F-Q4(a), 2024F-Q8, 2025F-Q2(b).

Pattern 3: Graph of the reciprocal function $y = 1/f(x)$ ★ Frequently tested in CF

Drawing rules (marking points in the key)

Feature of f(x)	Corresponding feature of 1/f(x)
Zero at $x = a$	Vertical asymptote $x = a$
$f \to \pm\infty$ as $x \to \pm\infty$	1/f has horizontal asymptote $y = 0$
Vertical asymptote of f at $x = b$	1/f has an open circle at (b, 0) (1/f is undefined at $x = b$, and 0 is removed from the range; 2025A-Q9(b))
Horizontal asymptote of f at $y = c$ ($c \neq 0$)	Horizontal asymptote of 1/f at $y = 1/c$ (in 2025A-Q9(b), $y = 2 \to y = 0.5$)
Local maximum (a, m), $m \neq 0$	Local minimum (a, 1/m) (and vice versa)
Point where $f(x) = \pm 1$	Point shared by f and 1/f (position unchanged)
$f > 0$ / $f < 0$	Sign unchanged
Open circle of f at (a, b), $b \neq 0$	Open circle of 1/f at (a, 1/b) (2018A-Q14(a)); if $b = 0$, it corresponds to a one-sided vertical asymptote (2022F-Q2)

Range: read the range from the graph of 1/f, and note any values that cannot be reached (2020F-Q5(b), 2024F-Q2(b), 2025A-Q9(b)).

Past questions: 2016F-Q8(a), 2018A-Q14(a), 2020F-Q5, 2022F-Q2(b), 2024F-Q2(b), 2025A-Q9(b).

Pattern 4: Absolute value functions

Three transformations

- $y = |f(x)|$: reflect the part below the x-axis above the x-axis.
- $y = f(|x|)$: keep the $x \geq 0$ part, then mirror it about the y-axis.
- $y = f(-|x|)$: keep the $x \leq 0$ part, then mirror it to the right (2023F-Q4(d)).

Common question types

- Determine the parameters of $y = a|x - b| + c$ from the graph (the vertex gives b and c, the gradient gives a) (2016A-Q12, 2017A-Q16(a)).
- $|f(x)| = d$ has exactly 4 solutions (2016A-Q12(b), $f(x) = -2|x - 3| + 5$): first draw the graph of $|f|$ (W shape), then cut it with the horizontal line $y = d$ to find the interval of d ($0 < d < 5$); check the endpoints $d = 0$ and $d = 5$ separately.
- Number-of-solutions function N(k): sweep $y = k$ across the graph of $|f(x)|$, and record the number of intersections section by section (2020A-Q21).
- The solution set of $|x + 1| = k - |x + a|$ is an interval $\Rightarrow$ rewrite as $|x + 1| + |x + a| = k$. The sum of two absolute values is constant $|a - 1|$ between $-a$ and -1 (flat bottom), and the interval of the flat bottom is the solution set $\Rightarrow a = 3$, $k = 2$ (2023F-Q4(e); the key says the two graphs coincide along a line segment).
- W-shaped graph $W_k(x) = |k/2 \cdot |x - k| - k|$: draw the inner layer first, then reflect; $\int_{k-2}^{k+2} W_k\, dx$ = area of one triangle between the two zeros $k \pm 2 = \frac{1}{2} \cdot 4 \cdot k = 2k$ (2024A-Q15).

Past questions: 2016A-Q12, 2016F-Q8(b), 2017A-Q16, 2018A-Q14(b)(c), 2019F-Q7, 2022F-Q2(a) ($|f(x)| = x$, the key requires excluding $x = 0$ and including $x = 2$), 2020A-Q21, 2023F-Q4(d)(e), 2024A-Q15, 2025F-Q2(c)(d).

Pattern 5: Rational functions — determining parameters from a graph

Approach: x-intercepts ⇒ factors in the numerator; vertical asymptotes ⇒ factors in the denominator; horizontal asymptote ⇒ ratio of leading coefficients (e.g. $a(x^2 - b)/((x + c)(x - d)) \to a$); a repeated root $(x - c)^2$ means the graph touches but does not cross the x-axis there, or the graph has the same sign on both sides of the asymptote; finally use the y-intercept or a known point to determine k.

Reasons must be written (the key usually awards 1 mark specifically for justification): e.g. "c = 3 since x = 3 is a vertical asymptote" (2023F-Q1, 2024F-Q5). When the denominator has a repeated factor $(x - c)^2$, the graph has the same sign on both sides of the asymptote (in 2023F-Q1 both sides tend to $-\infty$); when the numerator has a repeated factor, the graph touches the x-axis there but does not cross it. Oblique asymptote: polynomial division, f(x) = quotient + remainder/denominator ⇒ asymptote y = quotient (2024F-Q5(b), 2023A-Q18(a)). Rational function with a given axis of symmetry (2022A-Q15): $q(x) = a(x - 3)^2 + c$, a and c have the same sign (key: a, b > 0) is what guarantees no zeros; result $f = 2/(x^2 - 6x + 10)$.

Past questions: 2018F-Q4, 2020F-Q3, 2022A-Q15, 2023A-Q18, 2023F-Q1, 2024F-Q5, 2025A-Q9(a).

Pattern 6: Sketching rational functions

Checklist (1 mark each): x-intercept, y-intercept, vertical asymptote, horizontal or oblique asymptote, turning points (mark them if present; marks may be awarded even if the question does not explicitly ask, e.g. 2017F-Q5), curvature near asymptotes ($\to \pm\infty$ when approaching from the left and right sides). Technique: factorise first, and check for cancelled factors (holes); forms such as $f(x) = x - 1 - 3/(x + 1)$ let you read off the oblique asymptote y = x − 1 directly (2021F-Q4).

Past questions: 2017F-Q5, 2021F-Q4.